

Math Approach for LeetCode Problem

Smallest Integer Divisible by K

Hoai Linh DAO

November 25, 2025

Problem Statement

Let $a, b, c, d \in \mathbb{Z}$ with $b > 0$ and assume that

$$a \bmod b = r.$$

Prove that the following identity holds:

$$(ac + d) \bmod b = (rc + d) \bmod b.$$

Proof 1 (Algebraic / Ring-Theoretic)

Consider the canonical projection

$$\pi : \mathbb{Z} \rightarrow \mathbb{Z}_b, \quad \pi(x) = x \bmod b.$$

This map is a ring homomorphism, meaning it preserves addition and multiplication:

$$\pi(x + y) = \pi(x) + \pi(y), \quad \pi(xy) = \pi(x)\pi(y) \quad \text{for all } x, y \in \mathbb{Z}.$$

Let $f(x) = cx + d$ be an affine transformation. Applying π to both sides, we obtain

$$\pi(f(a)) = \pi(ca + d) = c\pi(a) + d.$$

Since $\pi(a) = a \bmod b = r$, it follows that

$$\pi(ca + d) = cr + d.$$

Therefore,

$$(ac + d) \bmod b = (rc + d) \bmod b,$$

Q.E.D.

Proof 2 (Elementary Number Theory)

From the assumption

$$a \bmod b = r,$$

there exists an integer q such that

$$a = bq + r, \quad 0 \leq r < b.$$

Now compute $ac + d$ using this representation:

$$ac + d = c(bq + r) + d = cbq + cr + d.$$

We examine this expression modulo b . The term cbq is divisible by b , so it does not affect the remainder:

$$ac + d \equiv cr + d \pmod{b}.$$

By the definition of the modulo operation, this is equivalent to

$$(ac + d) \bmod b = (cr + d) \bmod b.$$

This is precisely the identity we wanted to prove.

Q.E.D.